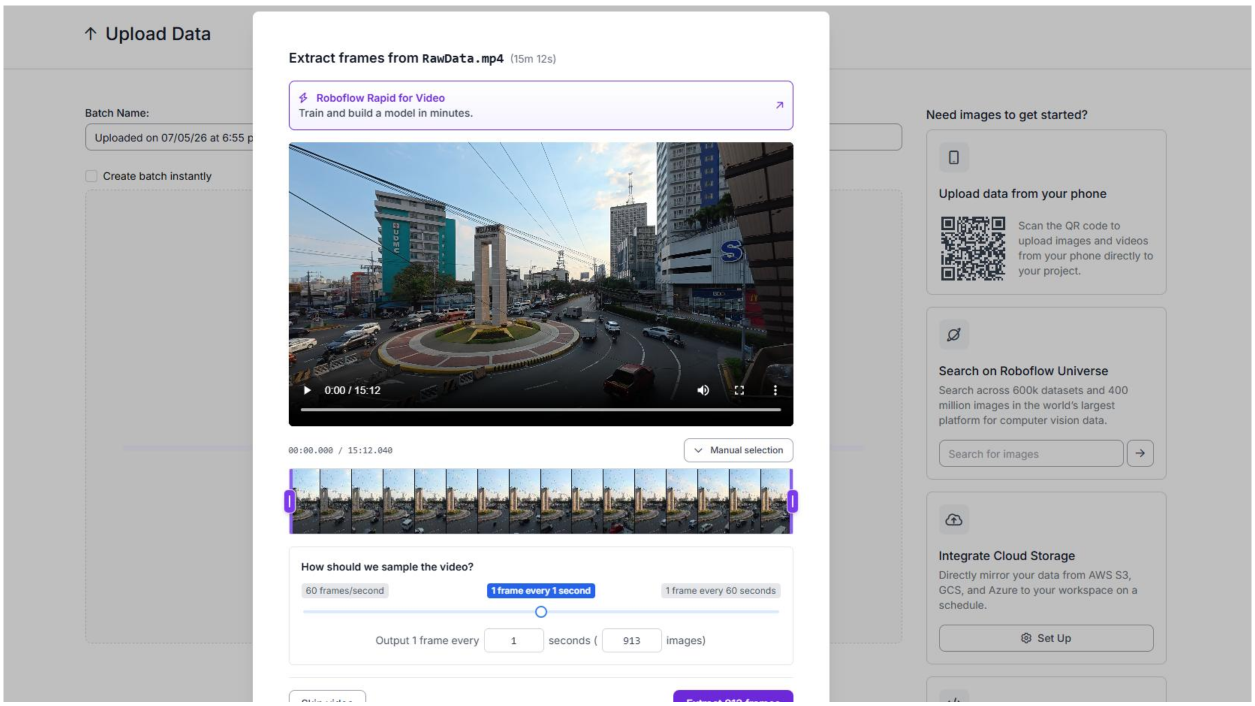
Shafer Arjun Tagorda

# Step-by-Step Annotation Process using Roboflow

## Step 1 – Creating Project:

A recording of Quezon City Roundabout was recorded within 15 minutes and inserted into Roboflow. 1 frame was sampled after every 1 second, creating 913 sample frames.




## Step 2 – Object Classifcation

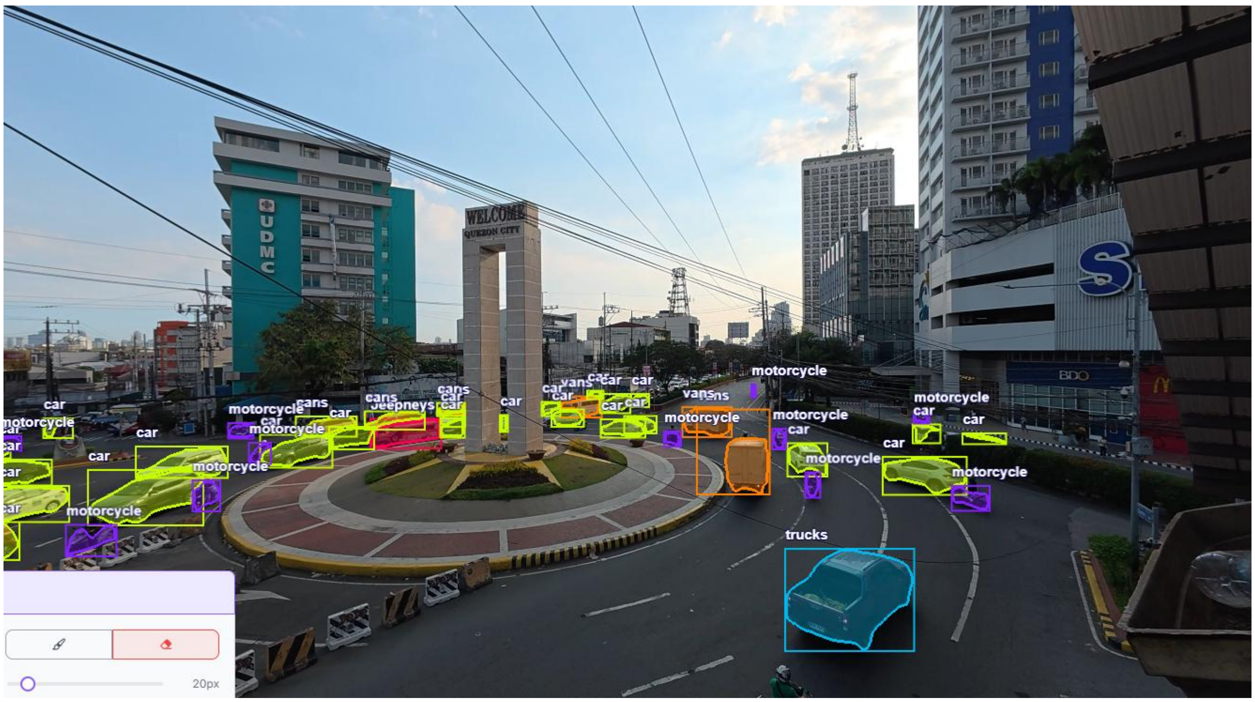
Six object classes were created:

- Jeepneys
- Bus
- Motorcycle
- Trucks

- Car
- Van

**Step 3 – Annotation**

In every frame, vehicles were labeled according to their object class.



# Quality Assurance Report

## Dataset Preparation and Conversion

The dataset was exported from Roboflow in YOLO format and validated for compatibility with Ultralytics YOLOv26. Because the downloaded version only contained training data, a reproducible 80/20 split was created programmatically using “random.seed(42)”, which produced 2,192 training images and 547 validation images. The final training structure was verified as:

- /train/images
- /train/labels
- /valid/images
- /valid/labels
- data_fixed.yaml

File allocation integrity was verified by matching image filenames with corresponding YOLO label files.

## Data Framing Size and Resolution

A 640x640 framing policy was used both in preprocessing and training to balance the computational cost and object visibility. This configuration is most suitable for baseline experiments and consistent with real-time detection constraint. Moderate augmentations were applied for task-appropriate road scenes:

- Flip: Horizontal
- Crop: 5%
- Rotation: ±7°
- Hue: ±3°
- Saturation: ±10%

- Brightness: ±12%
- Exposure: ±12%
- Blur: 2px
- Shear: ±2° Horizontal, ±2° Vertical

**Normalization**

Normalization requirements were already satisfied by the selected pipeline. The bounding boxes were normalized to YOLO's required [0,1] coordinate space. Pixel-level normalization was handled internally by Ultralytics preprocessing and pretrained model routines.

**Dataset Imbalance Assessment and Balancing Strategy**

Data imbalance should be identified using:

1. class frequency counts,
2. per-class image presence,
3. bounding-box size distributions, and
4. post-training confusion matrices and per-class recall/precision.

Potential imbalance effects include poor recall in minority classes and overprediction of majority classes. To address this, recommended balancing approaches are:

- targeted data collection for underrepresented classes,
- class-targeted augmentation,
- weighted sampling or class-aware losses, and
- class-specific threshold tuning during inference.

Model selection should prioritize not only overall mAP@50, but also balanced performance across all six classes.

**Metrics, Reproducibility, and Compliance**

Primary QA metrics are mAP@50, precision, recall, F1-score, and class-level confusion matrices. Reproducibility was maintained through fixed random seed usage, consistent class indexing (0–5), and explicit YAML configuration.
Overall, the dataset preparation pipeline is suitable for baseline YOLOv26 training after correction of the missing validation split. The current setup is methodologically sound for comparative model evaluation, while future improvements should emphasize class balancing and per-class error reduction.